

Real Analysis

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1 Introduction to Real Numbers

Properties of Rational Numbers: Rational numbers $\mathbb{Q}$ are numbers of the form $\frac{m}{n}$, where $m \in \mathbb{Z}$ and $n \in \mathbb{N}$.

- Equality:

$$\frac{m_1}{n_1} = \frac{m_2}{n_2} \Leftrightarrow m_1 n_2 = m_2 n_1$$

- Addition:

$$\frac{m_1}{n_1} + \frac{m_2}{n_2} = \frac{m_1 n_2 + m_2 n_1}{n_1 n_2}$$

- Multiplication:

$$\frac{m_1}{n_1} \frac{m_2}{n_2} = \frac{m_1 m_2}{n_1 n_2}$$

- Order:

$$\frac{m_1}{n_1} < \frac{m_2}{n_2} \Leftrightarrow m_1 n_2 < m_2 n_1$$

Show that multiplication is well-defined:

This mean if we have two representations of the same rational number

$$\frac{m_1}{n_1} = \frac{m_2}{n_2}$$

and likewise

$$\frac{p_1}{q_1} = \frac{p_2}{q_2}$$

then

$$\frac{m_1}{n_1} \frac{p_1}{q_1} = \frac{m_2}{n_2} \frac{p_2}{q_2}$$

Since we have that $m_1 n_2 = m_2 n_1$ and $p_1 q_2 = p_2 q_1$. Therefore

$$m_1 p_1 n_2 q_2 = (m_1 n_2)(p_1 q_2) = (m_2 n_1)(p_2 q_1) = m_2 p_2 n_1 q_1$$

Field: A Field $\mathbb{F}$ is a set with two operations that we denoting by '+' and '·'.

Additive properties:

- $x, y \in \mathbb{F}$, then $x + y \in \mathbb{F}$
- $x + y = y + x$
- $(x + y) + z = x + (y + z)$
- There exists an element $0 \in \mathbb{F}$ such that $0 + x = x$ for all $x \in \mathbb{F}$
- For all $x \in \mathbb{F}$ there exists an element, denoted by $(-x)$ such that $x + (-x) = 0$

Multiplication properties:

- $x, y \in \mathbb{F}$, then $xy \in \mathbb{F}$
- $xy = yx$
- $(xy)z = x(yz)$
- There exists an element, denoted by 1 such that $1x = x$ for all $x \in \mathbb{F}$
- For all $x \in \mathbb{F} \setminus \{0\}$ there exists an element, denoted by $\frac{1}{x}$ such that $x \frac{1}{x} = 1$

Axiom:

$$(x + y)z = xz + yz$$

Theorem: For any field 'zero' is unique.

Proof. Suppose there are two zero. Let denote as 0_1 and 0_2 . Then

$$0_1 + 0_2 = 0_2$$

Since 0_1 is a 'zero' and

$$0_1 + 0_2 = 0_1$$

Since 0_2 is a 'zero', therefore $0_1 = 0_2$ □

Ordered Set: An ordered set $\mathbb{S}$ is a set with a relation $<$ with the following properties:

- For an $x, y \in \mathbb{S}$, one of the following holds: $x < y$ or $x > y$ or $x = y$
- If $x, y, z \in \mathbb{S}$ with $x < y$ and $y < z$, then $x < z$

Ordered Field: An ordered Field is an ordered set that is also a Field and has the following two additional properties that chains the operations in the Field together with the ordering:

- If $x < y$, then $x + z < y + z$
- If $x > 0$, and $y > 0$, then $xy > 0$

Theorem: If $x < y$ and $z > 0$, then $xz < yz$

Proof. We need to show that $xz < yz$ or equivalent to $yz - xz > 0$.

Rewrite as $yz - xz = (y - x)z$. Since $y > x \Leftrightarrow y - x > 0$ and $z > 0$. by the properties of ordered field, $(y - x)z > 0$. □

Theorem: There does not exists a rational number x such that $x^2 = 2$.

Proof. Prove by contradiction. Suppose that there exists a rational number $x = \frac{m}{n}$ where $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, then

$$x^2 = \frac{m^2}{n^2} = \frac{m^2}{n^2} = 2$$

Assume that m and n does not have a common factor other than one. We have that $m^2 = 2n^2$, then 2 is a factor in m^2 . There exists m_1 such that $m = 2m_1$, where m_1 is an integer.

$$m^2 = 4m_1^2 = 2n^2 \Rightarrow n = 2m_1$$

Then n is also an even which contradicting that m and n have no common factor. This show that there was no rational number x with the property that $x^2 = 2$. □

Upper Bound: Suppose that $\mathbb{S}$ is an ordered set and A is a subset of $\mathbb{S}$, then M is an upper bound for A if for all $a \in A$ we have that $a \leq M$.

Completeness of Upper Bound: If a subset A of $\mathbb{R}$ has an upper bound, then A has a least upper bound

Least Upper Bound: Suppose that $\mathbb{S}$ is an ordered set and A is a subset that has an upper bound. We say that M is least upper bound for A if M is an upper bound for A and for any other upper bound M_1 we have that $M_1 \geq M$.

Theorem: There exists a complete ordered Field that contains $\mathbb{Q}$, denoted by $\mathbb{R}$

Theorem: $\sqrt{2} \in \mathbb{R}$

Proof. Let $A = (0, \sqrt{2}) \cap \mathbb{Q}$. That A consists all the positive rational numbers a so that $a^2 < 2$. Let x be the upper bound for A . Since $1 \in A$, then A is non-empty, and 2 is an upper bound of A . Note that $x \geq 1 > 0$ since it is an upper bound. Need to show that $x^2 = 2$

Show that $x^2 \leq 2$. Prove by contradiction that $x^2 > 2$. There exists $h > 0$, then make $x - h$ be the least upper bound.

$$(x - h)^2 = x^2 - 2hx + h^2 > x^2 - 2hx$$

$$x^2 - 2hx > 2 \Rightarrow h < \frac{2 - x^2}{2x}$$

Therefore $x - h$ is also upper bound for A which $\leq x$, that is contradicts that x is least upper bound.

Show that $x^2 \geq 2$. Prove by contradiction that $x^2 < 2$, we want to show that x is not upper bound for A . There exists $h > 0$, then make $(x + h) \in A$. If $0 < h < 1$ and h is small enough, $x + h \in A$.

$$(x + h)^2 = x^2 + 2hx + h^2 < x^2 + 2hx + h < 2$$

$$x^2 + 2hx + h < 2 \Rightarrow h < \frac{2 - x^2}{2x + 1}$$

Proved that $x + h \in A$, therefore $x^2 \geq 2$. Thus $x^2 = 2$

□

Theorem: $\mathbb{Q}$ is not complete.

Proof. If $\mathbb{Q}$ is complete, then $\sqrt{2} \in \mathbb{Q}$. By previous theorem, there is no rational number with the property that $x^2 = 2$.

□

Archimedean Property: For all $x \in \mathbb{R}$, there exists a natural $n \in \mathbb{N}$ such that $x < n$.

Proof. Prove by contradiction, suppose there exists an x such that all $n \in \mathbb{N}$, $n \leq x$.

That means x is the upper bound of $\mathbb{N}$. By least upper bound property: $\mathbb{N}$ is bounded subset of $\mathbb{R}$, then let y be the least upper bound of $\mathbb{N}$.

For all $n \in \mathbb{N}$ that $n \leq y$. Since $n + 1$ is also natural numbers, then we have that $n + 1 \leq y \Rightarrow n \leq y - 1$. Since n is any natural numbers, $y - 1$ would be an upper bound contradicting that y is least upper bound of $\mathbb{N}$.

□

Corollary: If $x < y$, then there exists a rational number $\frac{m}{n}$ such that

$$x < \frac{m}{n} < y$$

Proof. Set $\beta = \frac{1}{y-x}$, from Archimedean property there exists a natural number n with $n > \beta$, then $0 < \frac{1}{n} < \frac{1}{\beta}$. Let $m - 1$ be largest integer such that $m - 1 \leq xn$. Then

$$m > xn \Rightarrow x < \frac{m}{n}$$

$$0 < \frac{1}{n} < \frac{1}{\beta} = y - x \Rightarrow \frac{1 + xn}{n} < y$$

Since $m - 1$ is the largest integer, then substitute $m - 1$ into the inequality it remain valid.

$$\frac{1 + (m - 1)}{n} = \frac{m}{n} < y$$

Therefore we proved that $x < \frac{m}{n} < y$

□

Theorem: $\mathbb{R}$ is complete ordered Field that contains $\mathbb{Q}$.

Completeness is the property that every bounded non-empty subset has a least upper bound. Denoted by $\sup A$ the smallest upper bound of A .

Lower Bound: A non-empty subset A is said to have a lower bound if there exists $m \in \mathbb{S}$ such that for all $a \in A$ we have that $m \leq a$.

Greatest Lower Bound: A lower bound that is greater or equal to all other lower bounds, denoted by $\inf A$.

Theorem: There exists $a > 0$ such that $a^2 = 2$

Proof. Define a set A by

$$A = \{x \in \mathbb{R} | x > 0 \text{ and } x^2 < 2\}$$

Show that A is non-empty bounded subset and that $\alpha = \sup A$ has the property that $\alpha > 0, \alpha^2 = 2$

Observe that 2 is the upper bound for A , so A has least upper bound.

Show that $\alpha = 2^2$ in two parts:

Part 1: Show that $\alpha \leq 2^2$, Assume not, then $\alpha^2 > 2$. There exists α_0 with $0 < \alpha_0 < \alpha$ such that $\alpha_0^2 < 2$. Set

$$h = \frac{2 - \alpha^2}{4\alpha} \text{ and } \alpha_0 = \alpha - h$$

Then

$$\alpha_0^2 = (\alpha - h)^2 = \alpha^2 - 2h\alpha + h^2$$

Note that $0 < h < 1$

$$\alpha_0^2 = \alpha^2 - 2h\alpha + h^2 > \alpha^2 - 2h\alpha = \alpha^2 - \frac{2 - \alpha^2}{2} = \frac{\alpha^2}{2} - 1 \geq 2$$

Which means that for $\alpha_0 = \alpha - h$, α_0 also a upper bound and since $\alpha_0 < \alpha$, then α_0 is the least upper bound of A which shows the contradiction of the assumption. Therefore $\alpha^2 \leq 2$

Part 2: Show that $\alpha^2 \geq 2$. Assume not, then $\alpha^2 < 2$. If this was the case, then there exists an $\alpha_0 > \alpha$. Set

$$h = \frac{2 - \alpha^2}{4\alpha} \text{ and } \alpha_0 = \alpha + h$$

Then

$$\alpha_0^2 = \alpha^2 + 2h\alpha + h^2$$

Note that $0 < h < 1$, therefore $h^2 < h$. It follows that

$$\alpha_0^2 < \alpha^2 + 2h\alpha + h = \alpha^2 + h + \frac{2 - \alpha^2}{2}$$

Using that $h = \frac{2 - \alpha^2}{4\alpha} < \frac{2 - \alpha^2}{2}$, then

$$\alpha_0^2 < \alpha^2 + \frac{2 - \alpha^2}{2} + \frac{2 - \alpha^2}{2} = \alpha^2 + 2 - \alpha^2 = 2$$

Then $\alpha_0 \in A$ since $\alpha_0 > 0$ and $\alpha_0^2 < 2$. By assumption, $\alpha = \sup A$, but there exists a number that $\alpha_0 > \alpha$ and also an element of set A . Therefore there exists a contradiction implies that $\alpha^2 \geq 2$. We show that $\alpha^2 = 2$ as desired.

□

A sequence of real numbers is a function $f : \mathbb{N} \rightarrow \mathbb{R}$. Usually use the notation $a_n = f(n)$.

Example:

- $\sqrt{2}$ is the limit of $a_1 = 1, a_2 = 1.4, a_3 = 1.41, a_4 = 1.414$ etc
- $a_n = (-1)^n$. The sequence has no limit since a_n alternates between -1 and 1 .
- The sequence of $a_n = \frac{1}{n}$ has zero as its limit.

Limit: Let a_n be a sequence and a a real number. We say that a_n converges to a if for all $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that if $n \geq N$, then

$$|a_n - a| < \epsilon$$

If this is the case, then we say that a is the limit of the sequence and we say that the sequence is convergent. A sequence that is not convergent is said to be divergent.

Example: Define the sequence a_n as follows: Set $a_1 = 0.9, a_2 = 0.99, a_3 = 0.999$, etc. Prove that

$$\lim_{n \rightarrow \infty} a_n = 1$$

Proof. Show that for all $\epsilon > 0$, there exists N such that for $n \geq N$ we have that

$$|a_n - 1| < \epsilon$$

By the Archimedean property we can choose N such that $\frac{1}{N} < \epsilon$.

Therefore, for $n \geq N$, we have that

$$|a_n - 1| = 10^{-n} < \frac{1}{n} \leq \frac{1}{N} < \epsilon$$

□

Theorem: If a_n is convergent sequence, then the set $\{a_n\}$ is a bounded subset of $\mathbb{R}$.

Proof. Since a_n is convergent to a . In particularlly, if we chooses $\epsilon = 1$, then there exists N such that $n \leq N$

$$|a_n - a| < 1$$

Note also that the set $\{a_1, a_2, \dots, a_{N-1}\}$ is bounded, there exists $C \in \mathbb{R}$ such that for $n = 1, \dots, N-1$ we have that

$$|a_n| \leq C$$

By triangle inequality

$$|a_n| = |(a_n - a) + a| \leq |a_n - a| + |a| < 1 + |a|$$

Therefore, for all n

$$|a_n| \leq \max \{C, |a| + 1\}$$

□

Theorem: Suppose that a_n and b_n are convergent sequence with $\lim_{n \rightarrow \infty} a_n = a$ and $\lim_{n \rightarrow \infty} b_n = b$ and $C \in \mathbb{R}$, then

- $c_n = Ca_n$ is convergent and $\lim_{n \rightarrow \infty} c_n = Ca$.
- $c_n = a_n + b_n$ is convergent and $\lim_{n \rightarrow \infty} c_n = a + b$

Proof. If $c = 0$, then the claim is true. For $c \neq 0$, given that there exists an N such that if $n \geq N$, then

$$|a_n - a| < \frac{\epsilon}{|C|}$$

Multiplying both side by $|C|$ gives that

$$|Ca_n - Ca| < \epsilon, \quad \text{for } n \geq N$$

This show the first property. □

Proof. Since $a_n \rightarrow a$, given $\epsilon > 0$ we can find a N_a such that if $n \geq N_a$, then

$$|a_n - a| < \frac{\epsilon}{2}$$

Likewise since $b_n \rightarrow b$, we can find N_b such that $n \geq N_b$, then

$$|b_n - b| < \frac{\epsilon}{2}$$

Set $N = \max\{N_a, N_b\}$ and if $n \geq N$, then

$$|c_n - (a + b)| = |(a_n + b_n) - (a + b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b| = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

□

Proof. $\{a_n | n = 1, \dots\}$ is bounded, there exists $C \in \mathbb{R}$ such that $|a_n| \leq C$. Since $a_n \rightarrow a$, given $\epsilon > 0$, we can find N_a such that $n \geq N_a$, then

$$|a_n - a| < \frac{\epsilon}{2(|b| + 1)}$$

Likewise, since $b_n \rightarrow b$, then there exists N_b such that $n \geq N_b$, then

$$|b_n - b| < \frac{\epsilon}{2(C + 1)}$$

Since $c_n = a_n b_n$, then

$$|c_n - c| = |a_n b_n - ab| = |a_n b_n + a_n b - a_n b - ab| = |a_n(b_n - b) + b(a_n - a)| \leq |a_n| |b_n - b| + |b| |a_n - a|$$

For $n \geq \max\{N_a, N_b\}$.

$$|c_n - c| < C \frac{\epsilon}{2(C + 1)} + |b| \frac{\epsilon}{2(|b| + 1)} \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

□

Subsequence: A sequence a_n is a function $f : \mathbb{N} \rightarrow \mathbb{R}$ where we set $a_n = f(n)$. A subsequence b_n of a_n as a composition of functions $f \circ g$ where $g : \mathbb{N} \rightarrow \mathbb{N}$ is a strictly increasing function. So $b_n = f(g(n))$. Sometimes a subsequence of the sequence a_n also denoted by a_{n_k} .

Theorem: A sequence a_n is convergent with limit a if and only if all subsequence of a_n are also convergent with limit a .

Proof. Show that all subsequences of a_n are convergent with limit a , then the sequence a_n is convergent with limit a . Since a_n is a subsequence of itself, thus all subsequence of a_n convergent with limit a .

Show that any subsequence of a convergent sequence is convergent with the same limit.

Given $\epsilon > 0$ and there exists N such that $n \geq N$, then

$$|a_n - a| < \epsilon$$

For $k \geq N$, we have that $n_k \geq k \geq N$, therefore

$$|a_{n_k} - a| < \epsilon$$

□

Theorem: Monotone Convergent Theorem:

Increasing version: Suppose that a_n is a monotone increasing sequence that is bounded above, then there exists $A \in \mathbb{R}$ such that

$$a_1 \leq a_2 \leq \dots \leq A$$

then it is convergent with limit a , where

$$a = \sup \{a_n | n \in \mathbb{N}\}$$

Decreasing version: Suppose that a_n is a monotone decreasing sequence that is bounded below, then there exists $A \in \mathbb{R}$ such that

$$a_{n+1} \leq a_n$$

then it is convergent with limit a , where

$$a = \inf \{a_n | n \in \mathbb{N}\}$$

Cauchy Sequence: A sequence a_n is said to be a Cauchy sequence if for all $\epsilon > 0$, there exists an N such that if $m, n \geq N$, then

$$|a_n - a_m| < \epsilon$$

Cauchy Convergent Theorem: A sequence is convergent if and only if it is Cauchy sequence.

Proof. Let (x_k) be a sequence in $\mathbb{R}$.

- Suppose that (x_k) converges $(x_k) \rightarrow a$. Given $\epsilon > 0$, there exists $N \in \mathbb{Z}$ such that $k \geq N$

$$|x_k - a| < \epsilon$$

Then for $k, l \geq N$, we have

$$|x_k - x_l| = |x_k - a - x_l + a| \leq |x_k - a| + |x_l - a| = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus (x_k) is Cauchy sequence.

- Suppose that (x_k) is Cauchy sequence, we claim that (x_k) is bounded. Since (x_k) is Cauchy sequence, then there exists $N \in \mathbb{Z}$ $k, l \in \mathbb{Z}$ such that

$$|x_k - x_l| < 1$$

$$|x_k| = |x_k - x_l + x_l| \leq |x_k - x_l| + |x_l| = 1 + |x_l|$$

Therefore x_k is bounded by

$$b = \max \{|x_p|, |x_{p+1}|, \dots, |x_{l-1}|, 1 + |x_l|\}$$

Since (x_k) is bounded sequence, by bolzano-weirstrass theorem, the subsequence of x_k is convergent. Let (x_{k_j}) is subsequence of (x_k) .

$$\lim_{j \rightarrow \infty} x_{k_j} = a$$

Claim that the original sequence x_k converges with $\lim_{k \rightarrow \infty} x_k = a$. Given $\epsilon > 0$, since x_k is Cauchy sequence, there exists $N \in \mathbb{Z}$ such that $k, l \in \mathbb{Z} \Rightarrow |x_k - x_l| \leq \frac{\epsilon}{2}$. Since $x_{k_j} \rightarrow a$, there exists $N_1 \in \mathbb{Z}$ such that $j \geq N_1$. Choose an index $j \geq N_1$, then $k \geq N$.

$$|x_{k_j} - a| < \frac{\epsilon}{2}$$

$$|x_k - a| = |x_k - x_{k_j} + x_{k_j} - a| \leq |x_k - x_{k_j}| + |x_{k_j} - a| = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

□

Contracting Map: A map $T : \mathbb{R} \rightarrow \mathbb{R}$ such that there exists $c < 1$ so for all $x, y \in \mathbb{R}$, we have that

$$|T(x) - T(y)| \leq c|x - y|$$

For contracting map the fixed point is unique.

Proof. Suppose that x and y are two fixed point, show that $x = y$. We have

$$|x - y| = |T(x) - T(y)| \leq c|x - y|$$

Since $c < 1$, implies that $|x - y| = 0$, then $x = y$.

□

Contracting Mapping Theorem: Any contracting map has a fixed point.

Proof. Start with any x , where $x \in \mathbb{R}$ and form a sequence that $a_n = T^n(x)$. Then a_n is a Cauchy sequence and by Cauchy Convergence Theorem $a_n \rightarrow a$.

Show that a is a fixed point for T . Let $a_{n+1} = T(a_n)$, as $a_n \rightarrow a$, then $a_{n+1} \rightarrow a$.

$$|a_{n+1} - T(a)| = |T(a_n) - T(a)| \leq c|a_n - a|$$

Since $a_n \rightarrow a$, $|a_n - a| \rightarrow 0$. Therefore $|a_{n+1} - T(a)| \rightarrow 0$ implies that $a_{n+1} \rightarrow T(a)$ and also $a_{n+1} \rightarrow a$. Therefore $T(a) \rightarrow a$

□

Newton Method: Define a map $T(x) = x - \frac{f(x)}{f'(x)}$. Then

$$T'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{f'(x)^2} = 1 - \frac{f'(x)^2}{f'(x)^2} + \frac{f(x)f''(x)}{f'(x)^2} = f \frac{f''}{f'^2}$$

By MVT,

$$T(x) - T(y) = T'(\xi)(x - y)$$

If we knew that $T(\xi) \leq c < 1$, then

$$|T(x) - T(y)| = |T(\xi)| |x - y| \leq c |x - y|$$

Suppose that x_0 is a 'good' initial guess, that means $f(x_0)$ is extremely small.

Assume that $f' \neq 0$.

$$T(x) = x - \frac{f(x)}{f'(x)}$$

$$x_0 - \frac{f(x_0)}{f'(x_0)} = T(x_0) = x_0 \Rightarrow \frac{-f(x_0)}{f'(x_0)} = 0 \Rightarrow f(x_0) = 0.$$

Bolzano-Weirstrass Theorem: Any bounded sequence has a convergent subsequence.

Proof. Suppose that α_n is bounded sequence, assume $\alpha_n \in [0, 1]$

Defining subsequence α_{n_k} , either there are infinitely many n such that $\alpha_n \in [0, \frac{1}{2}]$ or there are infinitely many such α_n in $[\frac{1}{2}, 1]$. Assuming there are infinitely many in $[0, \frac{1}{2}]$. Set $\alpha_{n_1} = \alpha_1$, let α_{n_2} be the next α_n such that $\alpha_n \in [0, \frac{1}{2}]$, we have

$$\alpha_{n_1} \in [0, 1] \text{ and } \alpha_{n_2} \in \left[0, \frac{1}{2}\right]$$

Next, either infinitely many α_n lies in $[0, \frac{1}{4}]$, or infinitely many in $[\frac{1}{4}, \frac{1}{2}]$.

Assume that infinitely many lies in $[\frac{1}{4}, \frac{1}{2}]$. Pick $n > n_2$ such that $\alpha_n \in [\frac{1}{4}, \frac{1}{2}]$ and set $\alpha_{n_3} = \alpha_n$.

$$|\alpha_{n_1} - \alpha_{n_2}| \leq 2^{1-k}$$

where $k \in \mathbb{N}$. As $k \rightarrow \infty$, then $2^{1-k} \rightarrow 0$. It shows that the subsequence is Cauchy sequence, therefore the subsequence is convergent. □

Proof. Suppose that x_k be bounded sequence in $\mathbb{R}$. Choose $a, b \in \mathbb{R}$, with $a \leq x_k$ and $b \geq x_k$ for all k . Then $x_k \in [a, b]$ for all k , define a sequence of non-empty closed interval recursively.

$$I_0 = [a_0, b_0] = [a, b] = \left[a, \frac{a+b}{2}\right] \cup \left[\frac{a+b}{2}, b\right]$$

$$I_1 = [a_1, b_1] \text{ be equal one of two intervals } \left[a, \frac{a+b}{2}\right] \text{ or } \left[\frac{a+b}{2}, b\right]$$

There are infinitely many indices k with $x_k \in I_1$.

Suppose chosen interval $I_j = [a_j, b_j]$ with $b_j - a_j = \frac{1}{2^j}(b - a)$ for $1 \leq j \leq n$ such that $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots \supseteq I_n$. such that for each index j , there are infinitely many indices K with $x_k \in I_k$. Note that $I_n = [a_n, b_n] = \left[a_n, \frac{a_n+b_n}{2}\right] \cup \left[\frac{a_n+b_n}{2}, b_n\right]$. Obtained a sequence $(I_j)_{j \geq 0}$ of non-empty closed interval. By nested interval theorem $\bigcap_{j=0}^{\infty} I_j \neq \emptyset$. There exists c with $c \in I_n$ for $n \geq 0$.

Construct the subsequence of x_k with converges to c . For each j there exists infinitely many indices k with $x_k \in I_j$, we can construct a subsequence of (x_k) as follows. Choose k_0 such that $x_{k_0} \in I_0$, then choose $k_1 > k_0$ so that $x_{k_1} \in I_1$ and so on. Obtained a subsequence of x_k with $x_{k_j} \in I_j$ for all $j \geq 0$, claim that $x_{k_j} \rightarrow c$ as $j \rightarrow \infty$. Given $\epsilon > 0$, there exists N such that $\frac{1}{2^N}(b-a) < \epsilon$. For $j \leq m$ since $c \in [a, b]$ and $x_{k_j} \in [a_j, b_j]$, then

$$|x_{k_j} - c| = \max\{x_{k_j} - c, c - x_{k_j}\} \leq \frac{1}{2^j}(b-a) \leq \frac{1}{2^j}(b-a) < \epsilon$$

Thus $x_{k_j} \rightarrow c$ as $j \rightarrow \infty$. □

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at a point $x_0 \in \mathbb{R}$, if for all $\epsilon > 0$, there exists δ such that

$$|x_n - x_0| < \delta \Rightarrow |f(x_n) - f(x_0)| < \epsilon$$

A function is said to be continuous if it is continuous at all points in the domain.

Theorem: If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and x_n is a sequence, $x_n \rightarrow x_0$, then $f(x_n) \rightarrow f(x_0)$.

Proof. Since f is continuous, given $\epsilon > 0$ then there exists $\delta > 0$ such that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon$. Since $x_n \rightarrow x_0$, then there exists N such that $n \geq N$.

$$|x_n - x_0| < \delta \Rightarrow |f(x_n) - f(x_0)| < \epsilon$$

This shows that $f(x_n) \rightarrow f(x_0)$ □

Extreme Value Theorem: Let f be a continuous function on an interval $[a, b]$. The extreme value theorem says that the supremum and infimum are achieved. That is, there exist $x_M, x_m \in [a, b]$ such that

$$\sup_{[a,b]} f = f(x_M)$$

and

$$\inf_{[a,b]} f = f(x_m)$$

Proof. We will prove that the supremum is achieved. The proof for the infimum is analogous.

Let $M = \sup_{[a,b]} f$. By the definition of the supremum, there exists a sequence (x_n) in $[a, b]$ such that $\lim_{n \rightarrow \infty} f(x_n) = M$. Since (x_n) is contained in the bounded interval $[a, b]$, the sequence itself is bounded. By the Bolzano-Weierstrass Theorem, (x_n) has a convergent subsequence (x_{n_k}) that converges to some limit x_M .

Because $[a, b]$ is a closed interval, it contains all of its limit points. Therefore, we are guaranteed that $x_M \in [a, b]$. Since f is continuous on $[a, b]$, it is continuous at x_M . By the sequential characterization of continuity, we have:

$$\lim_{k \rightarrow \infty} f(x_{n_k}) = f(x_M)$$

On the other hand, since the original sequence of function values converges to M (i.e., $\lim_{n \rightarrow \infty} f(x_n) = M$), any of its subsequences must also converge to M . Therefore:

$$\lim_{k \rightarrow \infty} f(x_{n_k}) = M$$

By the uniqueness of limits, we conclude that $f(x_M) = M = \sup_{[a,b]} f$. □

Series: Suppose that a_n is a sequence, we can form s_n as follow:

$$s_1 = \alpha_1, s_2 = \alpha_1 + \alpha_2, \dots, s_n = \sum_{i=1}^{\infty} \alpha_i$$

Geometric Series: Suppose that $\alpha_n = c^n$ so the series is

$$s_n = \sum_{i=0}^n c^i$$

Convergent precisely when $|c| < 1$, moreover when $|c| < 1$ then the limit

$$\sum_{i=0}^n c^i = \frac{1}{1-c}$$

$$(1-c) \sum_{i=0}^n c^i = (1 + c + c^2 + \dots + c^n) - (c + c^2 + \dots + c^{n+1})$$

$$\sum_{i=0}^n c^i = \frac{1 - c^{n+1}}{1 - c}$$

The series is only convergent when $|c| < 1$, as $n \rightarrow \infty$, then $c^{n+1} \rightarrow 0$. Therefore

$$\sum_{i=0}^{\infty} c^i = \frac{1}{1-c}, \text{ where } |c| < 1.$$

Theorem: Geometric Convergent Theorem A series $\sum_{i=1}^{\infty} \alpha_i$ is convergent if the sequence s_n converges and if it do we also write $\sum_{i=1}^{\infty} \alpha_i$ for the limit.

Harmonic Series: For the harmonic series $\alpha_n = \frac{1}{n}$, so the series is $\sum_{i=1}^{\infty} \frac{1}{i}$. This series is divergent.

$$s_{2^n-1} \geq \frac{n}{2}$$

Proof. This is true for $n = 1$ as $s_1 = 1 \geq \frac{1}{2}$

Assume that also true for n , show that is true for $n + 1$.

$$s_{2^{n+1}-1} \geq s_{2^n-1} + \sum_{i=2^n}^{2^{n+1}-1} \frac{1}{i}$$

From 2^n to $2^{n+1} - 1$ there are 2^n numbers.

$$s_{2^{n+1}-1} \geq \frac{n}{2} + 2^n \frac{1}{2^{n+1}-1} \geq \frac{n}{2} + 2^n \frac{1}{2^{n+1}} = \frac{n}{2} + \frac{1}{2} = \frac{n+1}{2}$$

Since $\frac{n}{2} \rightarrow \infty$ it follow that the subsequence $s_{2^{n+1}-1}$ is divergent and therefore so is the original series. □

Absolutely Convergent: We say that series $\sum_{n=0}^{\infty} a_n$ is absolutely convergent if the series $\sum_{n=0}^{\infty} |a_n|$ is convergent.
Theorem: Absolutely convergent $\Rightarrow$ convergent.

Proof. Define that $s_n = \sum_{i=0}^n a_n$ and $\bar{s}_n = \sum_{i=0}^n |a_n|$. Let $m \leq n$. Use triangle inequality theorem

$$|s_n - s_m| = \left| \sum_{i=m+1}^n a_n \right| \leq |a_{m+1}| + \dots + |a_n| = |\bar{s}_n - \bar{s}_m|$$

There exists $N \in \mathbb{N}$ such that $n, m \geq N$, then $|\bar{s}_n - \bar{s}_m| = \bar{s}_n - \bar{s}_m < \epsilon$, therefore s is Cauchy sequence since the sequence is convergent. Therefore s_n is also Cauchy sequence, then s_n is convergent. $\square$

Counter Example: Alternatively Harmonic Series

$$\sum_{i=0}^{\infty} \frac{(-1)^n}{n}$$

is convergent but not absolutely convergent.

Theorem: A series of non-negative numbers

$$\sum_{i=0}^{\infty} a_i$$

where $a_n \geq 0$, is convergent if and only if the sequence s_n is bounded from above.

Proof. Since $a_n \geq 0$, then the sequence s_n is monotone nondecreasing since

$$s_{n+1} \geq s_n + a_{n+1} \geq s_n$$

The claim follows from the monotone convergent theorem, s_n is convergent if and only if s_n is bounded above. $\square$

Example: The series is defined as

$$\sum_{i=1}^{\infty} \frac{1}{i^2} \text{ is convergent}$$

Since the terms are non-negative, we only need to show there exists M such that for all n

$$s_n = \sum_{i=1}^n \frac{1}{i^2} \leq M$$

Claim that:

$$s_{2^n-1} \leq \sum_{i=0}^{n-1} \left(\frac{1}{2}\right)^i$$

Proof. We prove this by mathematical induction. For $n = 1$, we have:

$$s_1 = \frac{1}{1^2} \leq \left(\frac{1}{2}\right)^0 = 1$$

So the claim holds for $n = 1$. Assume that the claim holds for n , we will show that it also holds for $n + 1$.

$$s_{2^{n+1}-1} = s_{2^n-1} + \sum_{i=2^n}^{2^{n+1}-1} \frac{1}{i^2} \leq \sum_{i=0}^{n-1} \left(\frac{1}{2}\right)^i + 2^n \frac{1}{(2^n)^2} = \sum_{i=0}^{n-1} \left(\frac{1}{2}\right)^i + \frac{1}{2^n} = \sum_{i=0}^n \left(\frac{1}{2}\right)^i$$

This shows that the inequality also holds for $n + 1$. $\square$

Comparison Test(version 1): Suppose that a_n and b_n are two sequence with $0 \leq a_n \leq b_n$. If $\sum_{n=1}^{\infty} b_n$ is convergent, then $\sum_{n=1}^{\infty} a_n$ also convergent.

Example: The series $\sum_{n=1}^{\infty} \frac{n}{2^{-n}}$ is convergent. Set $a_n = \frac{2^{-n}}{n}$ and $b_n = 2^{-n}$, then $0 \leq a_n \leq b_n$. Since b_n is convergent, by comparison test, therefore the series $\sum_{n=1}^{\infty} \frac{2^{-n}}{n}$ is convergent.

Comparison Test(version 2): Suppose that a_n and b_n are two sequences with $b_n \neq 0$ and

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L \neq 0$$

The series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\sum_{n=1}^{\infty} b_n$ is convergent.

Example: The series $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$ is convergent. Let $a_n = \frac{1}{n^2-1}$ and $b_n = \frac{1}{n^2}$. Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2-1} = 1 \neq 0$$

Ans the series $\sum_{n=2}^{\infty} \frac{1}{n^2}$ is convergent, therefore the series $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$ also convergent.

Ratio Test: Let $a_n \geq 0$ and assume that $\frac{a_{n+1}}{a_n} = a$, if

- $a < 1$, then $\sum_{n=1}^{\infty} a_n$ is convergent.
- $a > 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent.
- $a = 1$, then it is inconclusive.

Example: $a_n = \frac{1}{n}$, in this case $\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \rightarrow 1$. The test is inconclusive but the series is divergent.

Exmaple: $a_n = \frac{1}{n^2}$, in this case $\frac{a_{n+1}}{a_n} = \frac{n^2}{n+1^2} \rightarrow 1$. So the test is inconclusive, but the series is convergent.

Root Test: Let $a_n > 0$ be a sequence of non-negative numbers. Suppose that $\lim_{n \rightarrow \infty} (a_n)^{\frac{1}{n}} = r$, if

- $r < 1$, then $\sum_{n=0}^{\infty} a_n$ is convergent.
- $r > 1$, then $\sum_{n=0}^{\infty} a_n$ is divergent.
- $r = 1$, then $\sum_{n=0}^{\infty} a_n$, then it is inconclusive.

Proof. Suppose $r < 1$, it follows that $r < r_0 < 1$, there exists $N \in \mathbb{N}$ such that $n \geq N$, then

$$(a_n)^{\frac{1}{n}} \leq r_0 \Rightarrow a_n \leq r_0^n$$

Since $r_0 < 1$, the series $\sum_{n=0}^{\infty} r_0^n$ is convergent. by the first version of comparison test, the series $\sum_{n=0}^{\infty} r^n$ is convergent. Suppose $r > 1$, it follows that $1 < r_0 < r$, there exists $N \in \mathbb{N}$ such that $n \geq N$, then

$$(a_n)^{\frac{1}{n}} \geq r_0 \Rightarrow a_n \geq r_0^n$$

Since $r_0 > 1$, the series $\sum_{n=0}^{\infty} r_0^n$ is divergent, implies that the original series also divergent. $\square$

Power Series:

•

$$\sum_{n=0}^{\infty} \frac{x^n}{n!}$$

•

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = \cos x$$

•

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sin x$$

Formal Definition: Let c_n be sequence, then $\sum_{n=0}^{\infty} c_n x^n$ is a power series.

Example: Consider the power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$. By ratio test, $a_n = \frac{x^n}{n!}$, we have

$$\frac{a_{n+1}}{a_n} = \frac{x^{n+1}n!}{x^n(n+1)!} = \frac{x}{n+1} \rightarrow 0$$

It follows that the power series is convergent for all x .

Let a_n be a sequence. If it not bounded from above, then we set $\limsup a_n$ to be ∞ .

Otherwise we will define a new sequence b_n from a_n as follows.

$$b_n = \sup \{a_n, a_{n+1}, \dots\}$$

Since we are assuming that a_n is bounded above, the b_n 's are real numbers and the sequence b_n is decreasing.

$$b_n = \sup \{a_n, a_{n+1}, \dots\} \geq \{\sup a_{n+1}, a_{n+2}, \dots\} = b_{n+1}$$

The sequence b_n is decreasing, if b_n is bounded below, let $b = \inf b_i$, otherwise it will converging to $-\infty$

$$\lim_{n \rightarrow \infty} \sup a_n = \lim_{n \rightarrow \infty} b_n = b$$

Likewise, a_n is any sequence bounded from below.

$$c_n = \inf \{a_n, a_{n+1}, \dots\} \leq \inf \{a_{n+1}, a_{n+2}, \dots\} = c_{n+1}$$

The sequence c_n is increasing, if c_n is bounded from above, let $c = \sup \{c_i\}$, otherwise it will converging to ∞

$$\lim_{n \rightarrow \infty} \inf a_n = \lim_{n \rightarrow \infty} c_n = c$$

Suppose that

$$\sum_{n=0}^{\infty} a_n x^n$$

is power series. For what x does the power series converges?

$$\lim_{n \rightarrow \infty} \sup \sqrt[n]{|a_n x^n|} = |x| \lim_{n \rightarrow \infty} \sup \sqrt[n]{|a_n|}$$

Theorem: Let f and g be two continuous function on $\mathbb{R}$ that agrees on all rational numbers, then $f = g$.

Theorem: All polynomial function are continuous.

Theorem: All ratioinal function are continuous.

Suppose that a_n and b_n are two convergent series of non-negative numbers $a_n, b_n \geq 0$. The product series $\sum_{n=0}^{\infty} c_n$, where $c_n = \sum_{i=0}^{\infty} a_i b_{n-i}$. Each $c_n \geq 0$ by Monotone Convergence Theorem, the series $\sum_{i=0}^{\infty} c_n$ is convergent if it bounded.

Theorem: If $\sum_{n=0}^{\infty} a_n$ and $\sum_{i=0}^{\infty} b_n$ of non-negative numbers $a_n, b_n \geq 0$, then the series $\sum_{n=0}^{\infty} c_n$ is convergent with the limit $\sum_{n=0}^{\infty} a_n \sum_{n=0}^{\infty} b_n$.

Proof. Denote $s_n^a = \sum_{i=0}^n a_i$ and $s_n^b = \sum_{i=0}^n b_i$ and $s_n^c = \sum_{i=0}^n c_i$. Then

$$|c_i| = \left| \sum_{i=0}^n a_i b_{n-i} \right| \leq \left| \sum_{i=0}^n |a_i| |b_{n-i}| \right|$$

$$S_n^c = \sum_{i=0}^n |c_i| \leq \sum_{k=0}^n \sum_{i=0}^k |a_i| |b_{k-i}| = \left(\sum_{i=0}^n |a_n| \right) \left(\sum_{i=0}^n |b_n| \right)$$

By previous theorem, $\sum_{i=0}^n |c_n|$ is convergent with the limit $\sum_{i=0}^n |a_i| \sum_{j=0}^n |b_j|$, therefore c_n is also absolutely convergent. $\square$

Theorem: $\mathbb{E}(x + y) = \mathbb{E}(x)\mathbb{E}(y)$

Proof. Assuming $x, y \geq 0$. Let $\mathbb{E}(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ and let $\mathbb{E}(y) = \sum_{n=0}^{\infty} \frac{y^n}{n!}$. Then

$$\mathbb{E}(x + y) = \sum_{n=0}^{\infty} \frac{(x + y)^n}{n!}$$

Set $c_n = \sum_{n=0}^{\infty} \frac{(x+y)^n}{n!}$. By binomial formula $(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^i y^{n-i}$ Then

$$c_n = \frac{\sum_{i=0}^n \binom{n}{i} x^i y^{n-i}}{n!} = \frac{\sum_{i=0}^n n! x^i y^{n-i}}{n! i! (n-i)!} = \sum_{i=0}^n \left(\frac{x^i}{i!} \right) \left(\frac{y^{n-i}}{(n-i)!} \right) = \sum_{i=0}^n a_i b_{n-i}$$

Where $a_i = \frac{x^i}{i!}$ and $b_{n-i} = \frac{y^{n-i}}{(n-i)!}$ $\square$

Chain Rule: Let $A, B \subset \mathbb{R}$ be open, let $f : A \rightarrow B$, let $g : B \rightarrow \mathbb{R}$ and $hg \circ f : A \rightarrow \mathbb{R}$. Let $a \in A$ and $b = f(a) \in B$. Suppose that f is differentiable at a and g is differentiable at b .

$$h'(a) = g'(f(a))f'(a)$$

Proof. Let $x \in A$ and $y = f(x) \in B$.

$$\begin{aligned} |h(x) - h(a) - g'(f(a))f'(a)(x - a)| &= |g(f(x)) - g(f(a)) - g'(f(a))f'(a)(x - a)| \\ &= |g(y) - g(b) - g'(b)f'(a)(x - a)| \\ &= |g(y) - g(b) - g'(b)(y - b) + g'(b)(y - b) + g'(b)f'(a)(x - a)| \\ &\leq |g(y) - g(b) - g'(b)(y - b)| + |g'(b)||f(x) - f(a) - f'(a)(x - a)| \end{aligned}$$

Also

$$\begin{aligned} |y - b| &= |f(x) - f(a) - f'(a)(x - a) + f'(a)(x - a)| \\ &= |f(x) - f(a) - f'(a)(x - a)| + |f'(a)||x - a| \end{aligned}$$

Let $\epsilon > 0$, since g is differentiable at b , given $\delta_0 > 0$ such that

$$|y - b| \leq \delta_0 \Rightarrow |g(y) - g(b) - g'(b)(y - b)| \leq \frac{\epsilon}{2(1 + |f'(a)|)} |y - b|$$

Since f is continuous at a . Given $\delta_1 > 0$ such that

$$|x - a| < \delta_1 \Rightarrow |f(x) - f(a)| \leq \delta_0$$

Since f is differentiable at a . Given $\delta_2 > 0$ and $\delta_3 > 0$ such that

$$\begin{aligned} |x - a| \leq \delta_2 &\Rightarrow |f(x) - f(a) - f'(a)(x - a)| \leq |x - a| \\ |x - a| \leq \delta_3 &\Rightarrow |f(x) - f(a) - f'(a)(x - a)| \leq \frac{\epsilon}{2(1 + |g'(b)|)} |x - a| \end{aligned}$$

Let $\delta = \min \delta_1, \delta_2, \delta_3$. Let $a \in A$ and $b = f(a) \in B$. $|x - a| \leq \delta$. Then

$$\begin{aligned} |h(x) - h(a) - h'(a)(x - a)| &\leq |g(x) + g(a) - g(f'(a))(f(x) - f(a))| + |g'(b)||f(x) - f(a) - f'(a)(x - a)| \\ &\leq \frac{\epsilon}{2(1 + |f'(a)|)} |y - b| + |g'(b)| \frac{\epsilon}{2(1 + |g'(b)|)} |1 + g'(b)||x - a| \\ &\leq \frac{\epsilon}{2(1 + |f'(a)|)} (|f(x) - f(a) - f'(a)(x - a)| + |f'(a)||x - a|) + \frac{\epsilon}{2} |x - a| \\ &\leq \frac{\epsilon}{2(1 + |f'(a)|)} (|x - a| + |f'(a)||x - a|) + \frac{\epsilon}{2} |x - a| \\ &\leq \frac{\epsilon}{2(1 + |f'(a)|)} |x - a| (1 + f'(a)) + \frac{\epsilon}{2} |x - a| \\ &\leq \frac{\epsilon}{2} |x - a| + \frac{\epsilon}{2} |x - a| \\ &\leq \epsilon |x - a| \end{aligned}$$

□

Lemma: If f is differentiable at x_0 , then f is continuous at x_0 .

Local Maximum: A function $f : \mathbb{R} \rightarrow \mathbb{R}$ has a local maximum at x_0 if there exists $\sigma > 0$ such that

$$f(x_0) = \max_{[x_0 - \delta, x_0 + \delta]} f = f(x_0)$$

Lemma: Let $f : [a, b] \rightarrow \mathbb{R}$ be a differentiable function and $x_0 \in (a, b)$ that f has a local maximum or minimum at x_0 , then

$$f'(x_0) = 0$$

Proof. Suppose that x_0 is local maximum, and local minimum is similar. Assume that all x is near to x_0

$$\frac{f(x) - f(x_0)}{x - x_0} \leq 0 \text{ for } x > x_0$$

$$\frac{f(x) - f(x_0)}{x - x_0} \geq 0 \text{ for } x < x_0$$

The limit is the same whether x converges to x_0 from the left or from the right it follows that $f'(x_0) = 0$. $\square$

Mean Value Theorem: Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable function, there exists a x_0 between a and b such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}$$

Proof. Suppose the function $g(x)$ is given by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a)$$

For $b \neq a$, then

$$g(a) = f(a) - \frac{f(b) - f(a)}{b - a}(a - a) = f(a)$$

$$g(b) = f(b) - \frac{f(b) - f(a)}{b - a}(b - a) = f(a)$$

By Rolle's theorem, there exists $x_0 \in (a, b)$ such that

$$g'(x_0) = \frac{g(b) - g(a)}{b - a} = \frac{f(a) - f(a)}{b - a} = 0$$

$\square$

Mean Value Theorem: Let $f, g : [a, b] \rightarrow \mathbb{R}$ be differentiable functions, then there exists a x_0 between a and b such that

$$f'(x_0)[g(b) - g(a)] = g'(x_0)[f(b) - f(a)]$$

observe that if $g(b) \neq g(a)$ and $g'(x_0) \neq 0$, then

$$\frac{f'(x_0)}{g'(x_0)} = \frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f(b) - f(a)}{b - a} \frac{b - a}{g(b) - g(a)}$$

Proof. Given the function $h(x)$

$$h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)]$$

$$h(a) = f(a)[g(b) - g(a)] - g(a)[f(b) - f(a)] = f(a)g(b) - g(a)f(b)$$

$$h(b) = f(b)[g(b) - g(a)] - g(b)[f(b) - f(a)] = f(a)g(b) - g(a)f(b)$$

Since $h(a) = h(b)$, by the Rolle's theorem, there exists $x_0 \in (a, b)$ such that $h'(x_0) = 0$

$$h'(x_0) = 0 = f'(x_0)[g(b) - g(a)] - g'(x_0)[f(b) - f(a)] \Rightarrow \frac{f'(x_0)}{g'(x_0)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$\square$

Taylor Expansion: Let $f : [a, b] \rightarrow \mathbb{R}$ be a function and k a positive integer. Assume that $f, f', \dots, f^{(k-1)}$ exists on $[a, b]$ and are continuous and that $f^{(k)}$ is defined on (a, b) , then there exists c between a and b such that

$$f(b) = f(a) + f'(a)(b - a) + \frac{f^{(2)}(a)}{2}(b - a)^2 + \dots + \frac{f^{(k-1)}(a)}{(k-1)!}(b - a)^{k-1} + \frac{f^{(k)}(c)}{k!}(b - a)^k$$

Proof. Define the Taylor polynomial by

$$P(x) = f(a) + f'(a)(x-a) + \dots + \frac{f^{(k-1)}(a)}{(k-1)!}(x-a)^{k-1}$$

and define a real number M such that

$$f(b) = P(b) - \frac{M}{k!}(b-a)^k$$

We want to show that $M = f^{(k)}(c)$ for some $c \in (a, b)$. Set

$$h(x) = f(x) - P(x) - \frac{M}{k!}(x-a)^k$$

$$h(a) = f(a) - P(a) = f(a) - f(a) = 0 \text{ and } h(b) = 0 \text{ by the definition of } M$$

By Rolle's theorem, there exists $c_1 \in (a, b)$ such that $h'(c_1) = 0$

$$h'(x) = f'(x) - P'(x) - \frac{M}{(k-1)!}(x-a)^{k-1}$$

For $x = a$

$$h'(a) = f'(a) - P'(a) = f'(a) - f'(a) = 0 = h'(c_1)$$

Since $h^{(i)}(a) = 0$ for all $i \geq 0$, Continue this process for k times, find some $c = c_k$ such that $h^{(k)}(c) = 0$

$$h^{(k)}(c) = 0 = f^{(k)}(c) - M$$

Therefore $f^{(k)}(c) = M$. Hence

$$f(b) = f(a) + f'(a)(b-a) + \frac{f^{(2)}(a)}{2}(b-a)^2 + \dots + \frac{f^{(k-1)}(a)}{(k-1)!}(b-a)^{k-1} + \frac{f^{(k)}(c)}{k!}(b-a)^k$$

□

Partition: Let $[a, b]$ be an interval. A partition $\mathbb{P}$ of the interval $[a, b]$ is a number of sub-divisoin x_i such that

$$a = x_0 < x_1 < \dots < x_n = b$$

The partition is then the sub-interval $[x_{i-1}, x_i]$. Set the $\Delta x_i = x_i - x_{i-1}$

Upper and lower sum: Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function and that $\mathbb{P} = x_i$ is a partition of the interval $[a, b]$. Define the upper and lower sums as follows. Set

$$M_i = \sup_{[x_{i-1}, x_i]} f$$

$$m_i = \inf_{[x_{i-1}, x_i]} f$$

and upper $U(f, \mathbb{P})$ and lower sums $L(f, \mathbb{P})$ by

$$U(f, \mathbb{P}) = \sum_{i=1}^n M_i \Delta x_i$$

$$L(f, \mathbb{P}) = \sum_{i=1}^{max} m_i \Delta x_i$$

Lemma: We always have that

$$U(f, \mathbb{P}) \geq L(f, \mathbb{P})$$

Sub-partition: Let $[a, b]$ be an interval and $\mathbb{P}_1$ and $\mathbb{P}_2$ two partition of the interval $[a, b]$. We say that $\mathbb{P}_2$ is a sub-partition of $\mathbb{P}_1$ if all the dividing points in $\mathbb{P}_1$ are also in $\mathbb{P}_2$

Lemma: Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a bounded funtion and that $\mathbb{P}_1$ is a partition of the interval $[a, b]$ and $\mathbb{P}_2$ is a refinement of $\mathbb{P}_1$, then

$$L(f, \mathbb{P}_1) \leq L(f, \mathbb{P}_2) \leq U(f, \mathbb{P}_2) \leq U(f, \mathbb{P}_1)$$

Proof. Suppose that $a = x_0 < x_1 < \dots < x_n = b$ are the dividing points for $\mathbb{P}_1$ and that between x_{i-1} and x_i there is an extre dividing point in $\mathbb{P}_2$ say y so $x_{i-1} < y < x_i$, then

$$\begin{aligned} \sup_{[x_{i-1}, y]} f &< M_i \\ \sup_{[y, x_i]} f &< M_i \\ \left[\sup_{[x_{i-1}, y]} f \right] (y - x_{i-1}) + \left[\sup_{[y, x_i]} f \right] (x_i - y) &\leq M_i \Delta x_i \end{aligned}$$

Therefore

$$U(f, \mathbb{P}_2) \leq U(f, \mathbb{P}_1)$$

□

Upper and lower integrals: Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function. Define the upper Riemann integral of f by

$$\int_a^{\bar{b}} f \, dx = \sup_{\mathbb{P}} U(f, \mathbb{P})$$

Define the lower Riemann integral of f by

$$\int_{\underline{a}}^b f \, dx = \inf_{\mathbb{P}} L(f, \mathbb{P})$$

Riemann Integrals: Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function, then we sya that f is Riemann integrable if

$$\int_a^{\bar{b}} f \, dx = \int_{\underline{a}}^b f \, dx$$

If the funtion is Riemann integrable, then the Riemann integral is

$$\int_a^b f \, dx = \int_a^{\bar{b}} f \, dx = \int_{\underline{a}}^b f \, dx$$

The Riemann integrable function is denoeted by $R([a, b])$

Lemma: Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a bouned function, then $f \in R([a, b])$ if and only if for all $\epsilon > 0$, there exists a partition $\mathbb{P}$ such that

$$U(f, \mathbb{P}) - L(f, \mathbb{P}) < \epsilon$$

Proof. Suppose that $f \in R([a, b])$, then

$$\sup_{\mathbb{P}} L(f, \mathbb{P}) = \int_{\underline{a}}^b f \, dx = \int_a^{\bar{b}} f \, dx = \inf_{\mathbb{P}} U(f, \mathbb{P})$$

That mean given that $\epsilon > 0$, there exists partition $\mathbb{P}_1$ and $\mathbb{P}_2$ such that

$$\int_a^b f \, dx - \frac{\epsilon}{2} < L(f, \mathbb{P}_1)$$

and

$$U(f, \mathbb{P}_2) \leq \int_a^b f \, dx + \frac{\epsilon}{2}$$

Let $\mathbb{P}$ be partition that has all dividing points of both $\mathbb{P}_1$ and $\mathbb{P}_2$. So $\mathbb{P}$ is a refinement of both $\mathbb{P}_1$ and $\mathbb{P}_2$

$$\int_a^b f \, dx - \frac{\epsilon}{2} < L(f, \mathbb{P}_1) \leq L(f, \mathbb{P}) \leq U(f, \mathbb{P}) \leq \int_a^b f \, dx + \frac{\epsilon}{2}$$

Prove the converse, suppose that for some $\epsilon > 0$, there exists a partition $\mathbb{P}$ such that

$$U(f, \mathbb{P}) - L(f, \mathbb{P}) < \epsilon$$

Since

$$L(f, \mathbb{P}) \leq \int_a^b f \, dx$$

and

$$U(f, \mathbb{P}) \geq \int_a^{\bar{b}} f \, dx$$

Then

$$\int_a^{\bar{b}} f \, dx - \int_a^b f \, dx \leq U(f, \mathbb{P}) - L(f, \mathbb{P}) < \epsilon$$

$f \in R([a, b])$ holds for all $\epsilon > 0$. □

Uniformly Continuous: Suppose that $f : I \rightarrow \mathbb{R}$ is a function, where I is an interval. We say that f is uniformly continuous if for all $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \epsilon \text{ if } |x - y| < \delta$$

Theorem: Any continuous function on $[a, b]$ is uniformly continuous.

Proof. Suppose not, then there exists $\epsilon > 0$ such that for all $n > 0$, there are x_n and y_n with

$$|x_n - y_n| < \frac{1}{n}$$

so that

$$|f(x_n) - f(y_n)| \geq \epsilon$$

Since the interval $[a, b]$ is compact, by Bolzano-Weirstrass theorem, there exists a subsequence of x_n say x_{n_k} so that $x_{n_k} \rightarrow x$ Since

$$|x - y_{n_k}| \leq |x - x_{n_k}| + |x_{n_k} - y_{n_k}|$$

We have that $y_{n_k} \rightarrow x$. Since f is continuous function we have that $f(x_{n_k}) \rightarrow f(x)$ and $f(y_{n_k}) \rightarrow f(x)$. Which contradict that

$$|f(x_{n_k}) - f(y_{n_k})| \geq \epsilon$$

□

Theorem: Any continuous function on $[a, b]$ is in $R([a, b])$

Proof. Given $\epsilon > 0$, since f is uniformly continuous by above theorem, then there exists $\delta > 0$ such that if $|x - y| < \delta$, then

$$|f(x) - f(y)| < \frac{\epsilon}{b - a}$$

Let P be a partition so that for all i we have that $\Delta x_i < \delta$, then on each interval of the partition of the form $[x_{i-1}, x_i]$, we have that

$$M_i - m_i < \frac{\epsilon}{b - a}$$

it follows that

$$U(f, P) - L(f, P) = \sum_i M_i \Delta_i - \sum_i m_i \Delta_i = \sum_i [M_i - m_i] \Delta x_i < \frac{\epsilon}{b-a} \sum_i \Delta x_i = \epsilon$$

□

Theorem:

- If $f \in R([a, b])$ and $c \in \mathbb{R}$, then $cf \in R([a, b])$ and

$$\int_a^b (cf) dx = c \int_a^b f dx$$

- If $f, g \in R([a, b])$, then $f + g \in R([a, b])$ and

$$\int_a^b (f + g) dx = \int_a^b f dx + \int_a^b g dx$$

- If $f, g \in R([a, b])$ and $f \leq g$, then

$$\int_a^b f dx \leq \int_a^b g dx$$

- If $f \in R([a, b])$ and $c \in (a, b)$, then $f \in R([a, c])$ and $f \in R([c, b])$ and

$$\int_a^c f dx + \int_c^b f dx = \int_a^b f dx$$

Proof. Prove of First Claim: Let $\mathbb{P}$ is a partiiton, then

$$U(cf, \mathbb{P}) = cU(f, \mathbb{P}) \text{ and } L(cf, \mathbb{P}) = cL(f, \mathbb{P})$$

□

Proof. Prove of Second Claim: Given $\epsilon > 0$, $\mathbb{P}_1$ and $\mathbb{P}_2$ be partition so that

$$U(f, \mathbb{P}_1) - L(f, \mathbb{P}_1) < \frac{\epsilon}{2}$$

$$L(g, \mathbb{P}_2) - L(g, \mathbb{P}_2) < \frac{\epsilon}{2}$$

Let $\mathbb{P}$ be refinement of $\mathbb{P}_1$ and $\mathbb{P}_2$. It follows that

$$U(f + g, \mathbb{P}) - L(f + g, \mathbb{P}) \leq U(f, \mathbb{P}_1) + U(g, \mathbb{P}_2) - L(f, \mathbb{P}_1) - L(g, \mathbb{P}_2) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

□

Proof. Prove of fourth claim: Given $\epsilon > 0$, since $f \in R([a, b])$, let $\mathbb{P}$ be the partition such that

$$U(f, \mathbb{P}) - L(f, \mathbb{P}) < \epsilon$$

Let $\mathbb{P}^*$ be the refinement in addition to all the dividing point of $\mathbb{P}$ also have c as dividing point.

$$\begin{aligned} U(f, \mathbb{P}^*) - L(f, \mathbb{P}^*) &< \epsilon \Rightarrow [U(f, \mathbb{P}^* \cap [a, c]) + U(f, \mathbb{P}^* \cap [c, b])] - [U(f, \mathbb{P} \cap [a, c]) + L(f, \mathbb{P} \cap [c, b])] \\ &\Rightarrow [U(f, \mathbb{P}^* \cap [a, c]) - L(f, \mathbb{P}^* \cap [a, c])] - [U(f, \mathbb{P}^* \cap [c, b]) - L(f, \mathbb{P}^* \cap [c, b])] \\ &< \epsilon \end{aligned}$$

□

Corollary: Suppose that $f, |f| \in R([a, b])$, then

$$\int_a^b f \, dx \leq \int_a^b |f| \, dx$$

Proof. Since $f \leq |f|$ and $-f \leq |f|$, so that

$$\int_a^b f \, dx \leq \int_a^b |f| \, dx \text{ and } \int_a^b -f \, dx = -\int_a^b f \, dx \leq \int_a^b |f| \, dx$$

□

Fundamental Theorem of Calculus: Let f be continuous function on $[a, b]$ and define F on $[a, b]$ by

$$F(x) = \int_x^a f(s) \, ds$$

The function $F(x)$ is differentiable with derivative of f .

Proof. Let $x_0 \in [a, b]$ and assume that $x > x_0$, then

$$F(x) - F(x_0) = \int_a^x f(s) \, ds - \int_a^{x_0} f(s) \, ds = \int_a^{x_0} f(s) \, ds + \int_{x_0}^x f(s) \, ds - \int_a^{x_0} f(s) \, ds = \int_{x_0}^x f(s) \, ds$$

$$\int_{x_0}^x f(s) \, ds \leq \int_{x_0}^x \sup_{[x_0, x]} f \, ds = \left(\sup_{[x_0, x]} f \right) (x - x_0)$$

Likewise

$$\int_{x_0}^x f(s) \, ds \geq \int_{x_0}^x \inf_{[x_0, x]} f \, ds = \left(\inf_{[x_0, x]} f \right) (x - x_0)$$

Therefore

$$\inf_{[x_0, x]} f \leq \frac{F(x) - F(x_0)}{x - x_0} \leq \sup_{[x_0, x]} f$$

Since f is continuous at x_0 , as $x \rightarrow x_0$, then both side convergent to $f(x_0)$. Similiarly as $x < x_0$, then

$$F(x) = F(x_0) - \int_x^{x_0} f(s) \, ds$$

$$\inf_{[x_0, x]} f \leq \frac{F(x) - F(x_0)}{x - x_0} \leq \sup_{[x_0, x]} f$$

Arguing as above gives the claim also in this case

□

Fundamental Theorem of Calculus: Suppose that $F : [a, b] \rightarrow \mathbb{R}$ is differentiable and that $F' = f \in R([a, b])$, then

$$F(b) - F(a) = \int_a^b f(s) \, ds$$

Proof. Since f is integrable, for all $\epsilon > 0$, there exists $\mathbb{P}$ as a partition of $[a, b]$ such that

$$U(f, \mathbb{P}) - L(f, \mathbb{P}) < \epsilon$$

$$U(f, \mathbb{P}) = \sum_i M_i(x_i - x_{i-1}) \text{ and } L(f, \mathbb{P}) = \sum_i m_i(x_i - x_{i-1})$$

By mean value theorem,

$$F(x_i) - F(x_{i-1}) = f(y_i)(x_i - x_{i-1}), \text{ for } y_i \in [x_{i-1}, x_i]$$

It follows that

$$L(f, \mathbb{P}) \leq \sum_i^n [F(x_i) - F(x_{i-1})] \leq U(f, \mathbb{P})$$

Therefore

$$F(b) - F(a) = \sum_i^n [F(x_i) - F(x_{i-1})] = \int_a^b f(s) \, ds$$

□

Pointwise Convergence: Suppose that f_n is a sequence of functions on an interval I , then we say that f_n convergence pointwise to a function f if for all x we have that $f_n(x) \rightarrow f(x)$.

Uniform Convergence: Suppose that f_n is a sequence of functions on an interval I , then we say that f_n converges uniformly to a function f if for all $\epsilon > 0$, there exists an N such that $n \geq N$, then for all x

$$|f(x) - f_n(x)| < \epsilon$$